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Standard dynamic model and control of a two-wheeled inverted pendulum robot on inclined planes

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Abstract
Recently, there has been significant attention on the dynamic model of two-wheeled inverted pendulum robots (TWIPRs) on horizontal planes, leaving an inadequacy in understanding their behavior on inclined planes. This paper aims to address this gap by establishing a standard three-degree-of-freedom (3-DOF) nonlinear model for TWIPRs on inclined planes, particularly considering the complex coupling among yaw motion, forward motion, and body balance. First, to accommodate varying incline angles, multiple coordinate systems are introduced, including one aligned with the inclined plane. Second, through the analysis of kinematics and dynamics, the Lagrangian method is employed to derive the standard model, incorporating the effects of yaw motion on potential energy, slope angles, and pitch angles. Third, a sliding mode control strategy is applied in the model to guide TWIPRs along predefined paths on inclined planes for maintaining balance. Finally, numerical simulations demonstrate the model's effectiveness and show its superior accuracy in trajectory tracking on inclined planes compared to the previous models.

Keywords Two-wheeled inverted pendulum robot · Inclined plane dynamics · Sliding mode control · Lagrangian method

1 Introduction

TWIPRs have attracted significant attention due to their unique physical characteristics and advanced control methods. As a classical inverted pendulum system, the high nonlinearity and inherent instability of TWIPR make it an ideal platform for control theory and method validation [1–5]. The simple but robust mechanical design, consisting of a robot body with two collinearly mounted wheels and a centrally positioned chassis, offers advantages such as low cost, high flexibility, and easy maneuverability. These features enable TWIPRs to excel in diverse applications such as automated transportation, household services, and rescue missions [6, 7]. Moreover, in confined spaces and complex terrains, the high maneuverability and flexibility of TWIPR significantly enhance its practical value, making TWIPR an indispensable tool [8]. Therefore, studying TWIPR is of great significance for advancing control theory and expanding its practical applications.

Understanding the dynamic model of TWIPRs is crucial to fully exploit their potential. Early on, Yuta et al. developed a dynamic model of TWIPR for balance and forward motion on a horizontal plane, allowing the experimental platform "Yamabico Kurara" to maintain balance while moving at a specified speed on a flat surface[9]. Subsequently, Grasser et al. developed a 3-DOF linear model of the motion of TWIPRs on a plane to address the yaw motion problem, by decoupling the system into two subsystems and designing controllers for operator command tracking [10]. However, these models are too simple to adequately describe the coupling between balance, forward motion, and yaw motion in TWIPRs. This limitation reduces their kinematic and dynamic performance, especially during yaw motion. As TWIPRs developed, a reliable model has become increasingly important for controller design. For example, in [11], Kim et al. improved the dynamic model of TWIPRs using the Lagrangian method and Kane's method, enhancing reliability during yaw motion. Subsequently, Ghaffari et al. identified a previously overlooked term and further refined the model, significantly improving

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the dynamic performance of TWIPR [12]. With continued advancements, the planar model and its associated control systems have become highly refined, on the horizontal plane, providing a robust foundation for balance, forward motion, and yaw motion[13].

On the other hand, model and stability control of TWIPRs on inclined planes remain a significant challenge. Traditional planar models often fail to capture the complex dynamics of TWIPRs in this situation. Thus, it is necessary to research the nonlinear dynamic model and control strategies for TWIPRs on inclined planes to enhance their performance. In [14, 15], a 2-DOF model was developed to describe the linear motion of TWIPRs on an inclined plane, which estimated the incline angle and maintained balance by a disturbance observer. Additionally, a dynamic model was proposed in [16] for TWIPRs operation on uneven, inclined, and flat terrains, exploring the impact of terrain inclination on performance. However, these studies do not address the yaw motion of TWIPRs on inclined planes. In [17], Peng et al. considered yaw motion in their model, allowing TWIPRs to handle small-angle yaw movements. Similarly, Kim et al. considered the dynamic characteristics of TWIPRs on inclined planes, combining a planar model with a disturbance observer to support yaw motion [18]. Nonetheless, the models mentioned before for motion on inclined planes are relatively simple and have been designed to accommodate the limited processing ability of available hardware. Despite their practical applicability, these models have inherent limitations. These 3-DOF models struggle to accurately describe the complex coupling relationships between yaw motion, balance, and forward motion, as well as to characterize the effects of inclined planes on movement. These limitations restrict the use of the control methods, thereby reducing the agility and accuracy of the TWIPRs on inclined surfaces.

As is well known, sliding mode control (SMC), being one of the most classical control methods, has been widely adopted due to its excellent robustness [19, 20]. Yue et al. introduced the zero-dynamics method into the SMC design, dividing the TWIPR system into three subsystems and designing SMC for them [21]. Xu et al. proposed an Integral Sliding-Mode Control (ISMC) method to regulate and control TWIPRs under matched and unmatched uncertainties [22]. Dao et al. combined the online adaptive reinforcement learning control and the integral sliding mode control to improve the tracking performance of a wheeled inverted pendulum [23]. Huang et al. designed an SMC controller based on a high-order disturbance observer, effectively reducing the chattering phenomenon while ensuring control performance [24]. Due to their robustness to external disturbances, SMC controllers have been widely used in other TWIPR applications [25–28]. However, to achieve optimal performance, an effective control strategy must be based on an accurate model.

In this work, our aim is to develop a comprehensive 3-DOF nonlinear model for TWIPRs on inclined planes. This model considers the complex coupling between yaw motion, forward motion, and body balance, and is generalized for different incline angles using multiple coordinate systems. Moreover, we introduce an SMC strategy based on this model, enabling TWIPRs to achieve precise trajectory tracking on inclined planes. Numerical simulations demonstrate the model's superior accuracy and robustness compared to the previous models, particularly in handling the dynamic challenges posed by inclined surfaces.

This paper is structured as follows. Section 2 provides preliminaries that help explain kinematics, kinetic energy, and potential energy. In addition, the standard model, generated using a Lagrangian approach, is also presented. Section 3 establishes zero-dynamics controllers that depend on the model. Section 4 shows the principal simulation results, verifying the effectiveness of the standard model and its controller. Finally, Sect. 5 concludes the observations of this work. The aforementioned symbols, not thoroughly illustrated in Sect. 3, are listed in the appendix.

2 The standard model of TWIPR

In this section, we aim to establish the dynamic equations of TWIPR with 3-DOF on inclined planes. Initially, we introduce the essential assumptions and parameters in the preliminary part. Then, we establish a coordinate system to obtain the Kinematic constraints. Finally, the Lagrangian approach is applied to obtain the standard model through the expression of TWIPR's kinetic energy and potential energy.

2.1 Preliminary

In this work, four assumptions have been imposed on the TWIPR and the terrain: Bulleted lists look like this:

- A1: The inclined terrain has a flat surface;
- A2: Each part of the TWIPR is symmetrical and ideal rigid body;
- A3: No slip on the ground;
- A4: No friction resistance inside the TWIPR or from the surface.

From a mathematical modeling perspective, TWIPR consists of three rigid parts which are two wheels connected by the chassis and the body which acts like an inverted pendulum. The schematic of the TWIPR is displayed in Fig. 1, and the important variables and parameters are listed in Table. 1.

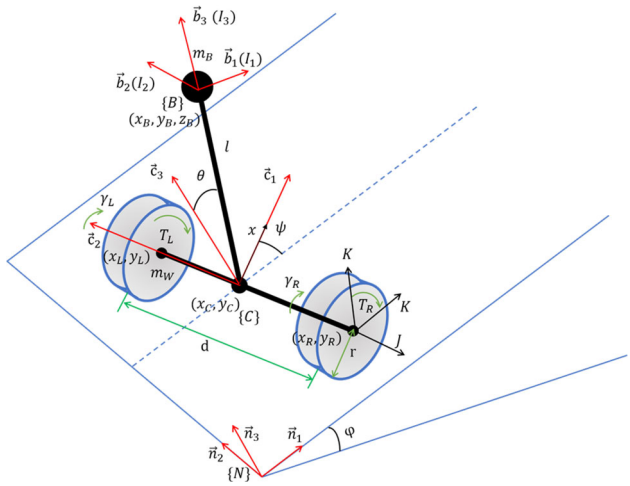

Fig. 1 Coordinate system, rotational axis of moment of inertia, parameters and variables of the TWIPR

Table 1 The parameters and variables of a TWIPR

Symbol	Unit	Definition
m_B	kg	Mass of the robot body
m_W	kg	Mass of the wheel
K, J	kg·m^2	Inertia moment of each wheel
I_1, I_2, I_3	kg·m^2	Inertia moment of the robot body
d	m	Distance between two wheels
r	m	Radius of each wheel
l	m	Distance between center of chassis and body center of gravity
g	m/s^2	Acceleration of gravity
φ	rad	Angle of slope
T_L, T_R	N·m	Torques acting on the left and right wheels provided by the wheel motors
θ	rad	Tilt angle of the robot body
ψ	rad	Rotational angle of the robot
x	m	Displacement of the robot along the direction of longitudinal velocity
γ_L, γ_R	rad	Rotational angles of the left and right wheels

2.2 Kinematics

If the coordinate system is established on a horizontal plane, the resulting inclined plane model will become overly complex. This complexity has also limited the improvement of TWIPR models in most previous works. However, establishing the coordinate system directly on the inclined plane can make the modeling process more detailed, a level of detail that has not been achieved in earlier works on inclined plane models.

When a TWIPR working with 3-DOF on inclined planes, using the coordinate system which is established on a horizontal plane to describe the state of TWIPR is complicated, such as all the velocities and the x-y plane are not parallel to the inclined surface. Therefore, rectangular coordinates are set on the inclined surface, denoted as {N}, {C}, and {B}in Fig. 1. The origin of the {N} frame is at the initial position of the robot. The $\vec{n}_1$-axis rises in the direction of steepest ascent on the slope, the $\vec{n}_2$-axis follows the direction of potential energy variation, and the $\vec{n}_3$-axis is perpendicular to the inclined terrain. The chassis frame {C} is attached at the center of the wheel axis, with the $\vec{c}_1$-axis aligned with the direction of the TWIPR's longitudinal velocity. The body frame {B} has its origin located at the body's center of mass, and its $\vec{b}_1$-axis points outward in the direction of the body. With these frames, the TWIPR's straight motion, yaw motion, and body's pitch motion can be easily described.

To describe each part's kinematics of TWIPR in detail, 13 generalized coordinates noted in Fig. 1 are selected: x_B, y_B, z_B, x_L, y_L, x_R, y_R, x_C, y_C, γ_L, γ_R, θ, ψ, where x_B, y_B, z_B denote the coordinates of the body in frame {N}, and x_L,y_L,x_R, y_R,x_C,y_C denote the coordinates of the wheels and chassis in frame {N}.

Due to the connections of TWIPR's wheels and the body, we can derive geometric constrains:

$$\begin{aligned} &x_B = x_C + l sin\theta cos\psi,\ y_B = y_C + l sin\theta sin\psi,\ z_B = l cos\theta, \\ &x_L = x_C - \frac{d}{2} sin\psi, \qquad y_L = y_C + \frac{d}{2} cos\psi, \\ &x_R = x_C + \frac{d}{2} sin\psi, \qquad y_R = y_C - \frac{d}{2} cos\psi. \end{aligned} \tag{1}$$

As assumption A3, there is no velocity or instantaneous displacement along the $\vec{c}_1$-axis. It means the linear velocity of each wheel is determined by its rotational velocity and radium as shown in Fig. 2. Thus, the difference constraints can be derived:

$$\begin{aligned} &\dot{x}_L sin\psi = \dot{y}_L cos\psi, \quad \dot{x}_R sin\psi = \dot{y}_R cos\psi, \\ &\dot{x}_L cos\psi + \dot{y}_L sin\psi = r\dot{\gamma}_L, \quad \dot{x}_R cos\psi + \dot{y}_R sin\psi = r\dot{\gamma}_R. \end{aligned} \tag{2}$$

And the velocity of the chassis is determined by velocities of two wheels:

$$\begin{aligned} \dot{x}_C &= \frac{\dot{x}_L + \dot{x}_R}{2}, \\ \dot{y}_C &= \frac{\dot{y}_L + \dot{y}_R}{2}, \\ \dot{x} &= \frac{r\dot{\gamma}_L + r\dot{\gamma}_R}{2}. \end{aligned} \tag{3}$$

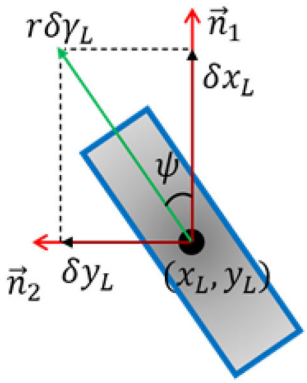

Fig. 2 Differential constraints of left wheel

By using Eqs. (1) and (3), the equation Eq. (2) can be rewritten as:

$$\begin{aligned} &\dot{x}_C sin\psi = \dot{y}_C cos\psi, \\ &\dot{x}_C cos\psi + \dot{y}_C sin\psi - \frac{d}{2}\dot{\psi} - r\dot{\gamma}_L = 0, \\ &\dot{x}_C cos\psi + \dot{y}_C sin\psi + \frac{d}{2}\dot{\psi} - r\dot{\gamma}_R = 0. \end{aligned} \tag{4}$$

Summing the last two equations in Eq. (2) with Eq. (3), the longitudinal velocity $\dot{x}$ can be decomposed along the frame {N} as:

$$\dot{x} = \dot{x}_C cos\psi + \dot{y}_C sin\psi. \tag{5}$$

Then, from Eq. (4), we have the rotational velocity of left and right wheel:

$$\dot{\gamma}_L = \frac{\dot{x} - \frac{d}{2}\dot{\psi}}{r}, \dot{\gamma}_R = \frac{\dot{x} + \frac{d}{2}\dot{\psi}}{r}. \tag{6}$$

2.3 Lagrangian approach

In order to apply the Lagrangian approach, we are supposed to calculate the kinetic energy and potential energy. Firstly, the linear velocities of each part are given by differentiating both side of Eq. (1):

$$\begin{aligned} v_B &= \left(\dot{x}_C + l\dot{\theta}cos\theta cos\psi - l\dot{\psi}sin\theta sin\psi\right)\vec{n}_1 \\ &\quad + \left(\dot{y}_C + l\dot{\theta}cos\theta sin\psi + l\dot{\psi}sin\theta cos\psi\right)\vec{n}_2 \\ &\quad - (l\dot{\theta}sin\theta)\vec{n}_3, \\ v_L &= \left(\dot{x}_C - \frac{d}{2}\dot{\psi}cos\psi\right)\vec{n}_1 + \left(\dot{y}_C - \frac{d}{2}\dot{\psi}sin\psi\right)\vec{n}_2, \\ v_R &= \left(\dot{x}_C + \frac{d}{2}\dot{\psi}cos\psi\right)\vec{n}_1 + \left(\dot{y}_C + \frac{d}{2}\dot{\psi}sin\psi\right)\vec{n}_2, \end{aligned} \tag{7}$$

where v_B,v_L,v_R are the body, left wheel and right wheel linear speeds.

By using Eq. (7) the translational kinetic energy of the TWIPR can be expressed as:

$$T_{trans} = \frac{1}{2}m_W(v_L \cdot v_L) + \frac{1}{2}m_W(v_R \cdot v_R) + \frac{1}{2}m_B(v_B \cdot v_B). \tag{8}$$

As shown in Fig. 1, the wheels' angular velocities come from the chassis rotation on $\vec{c}_3$-axis and wheel rotation on $\vec{c}_2$-axis. Similarly, the angular velocities of body come from the body rotation on $\vec{b}_2$-axis and the chassis rotation on $\vec{c}_3$-axis. Then we have:

$$\begin{aligned} \omega_L &= \omega_C + \dot{\gamma}_L\vec{c}_2 = \dot{\psi}\vec{c}_3 + \frac{\dot{x} - \frac{d}{2}\dot{\psi}}{r}\vec{c}_2, \\ \omega_R &= \omega_C + \dot{\gamma}_R\vec{c}_2 = \dot{\psi}\vec{c}_3 + \frac{\dot{x} + \frac{d}{2}\dot{\psi}}{r}\vec{c}_2, \\ \omega_B &= \omega_C + \dot{\theta}\vec{b}_2 = \dot{\psi}\vec{c}_3 + \dot{\theta}\vec{b}_2 \\ &= (-\dot{\psi}sin\theta)\vec{b}_1 + \dot{\theta}\vec{b}_2 + (\dot{\psi}cos\theta)\vec{b}_3, \end{aligned} \tag{9}$$

where ω_B,ω_L,ω_R are the body, left wheel and right wheel angular speeds.

With Eq. (9), the rotational kinetic energy can be calculated as:

$$T_{rot} = \frac{1}{2}\omega_L^T \cdot I_L \cdot \omega_L + \frac{1}{2}\omega_R^T \cdot I_R \cdot \omega_R + \frac{1}{2}\omega_B^T \cdot I_B \cdot \omega_B. \tag{10}$$

where $I_L = I_R = diag\{K, J, K\}$, $I_B = diag\{I_1, I_2, I_3\}$ as the assumption A2.

Then, the kinetic energy can be calculated by:

$$T = T_{rot} + T_{trans}. \tag{11}$$

Due to the TWIPR works on inclined terrain, the {N} and {C} frame is established on the slope. As shown in Fig. 3, it makes the velocities and kinetic energy calculation similar to the model on horizontal terrain, but makes the potential energy is complicated to express. Previous work did not thoroughly analyze potential energy, especially when the yaw angle is nonzero, leading to errors in some modeling efforts. To better analyze the potential energy, it is divided into four parts as noted in Fig. 4.

The base potential energy V_1 is caused by the displacement of TWIPR along the $\vec{n}_1$-axis (the displacement along the $\vec{n}_2$-axis makes no difference to the potential energy), which is noted in Fig. 4 as h_1. Therefore, we have:

$$V_1 = (m_B + 2m_W)g\sin\varphi x_C. \tag{12}$$

The potential energy V_2 comes from the wheel is invariant because the assumption A2 which means the wheel will not

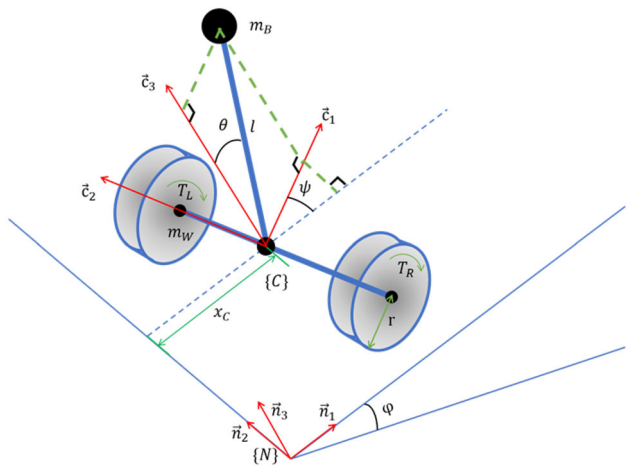

Fig. 3 Full view of the TWIPR potential energy calculation

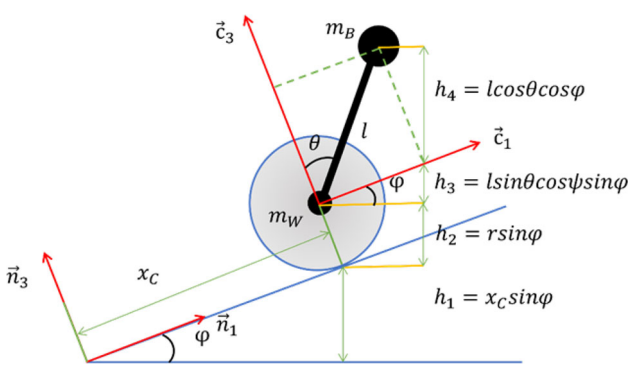

Fig. 4 Side view of TWIPR potential energy calculation

deform:

$$V_2 = (m_B + 2m_W)gsin\varphi r. \tag{13}$$

And the potential energy from the pitch motion of body is split into two parts: one is the projection of the body on the $\vec{c}_3$-axis, which noted as h_4 in Fig. 4; the other is the projection of the body on the $\vec{c}_1$-axis, which noted as h_3 in Fig. 4. The V_3 and V_4 can be expressed as follows:

$$\begin{aligned} V_3 &= m_B glsin\theta cos\psi sin\varphi, \\ V_4 &= m_B glcos\theta cos\varphi. \end{aligned} \tag{14}$$

Besides, we can calculate the potential energy by the angle between the body and the straight line z of the vertical horizontal plane. However, due to the $\vec{c}_1$-axis, $\vec{b}_3$-axis and the straight line z are not always in the same plane, but constituting the three lateral edges of a triangular pyramid in Fig. 5, it is necessary to indicate that the angle cannot be calculated directly by adding the slope angle φ and the tilt angle θ.

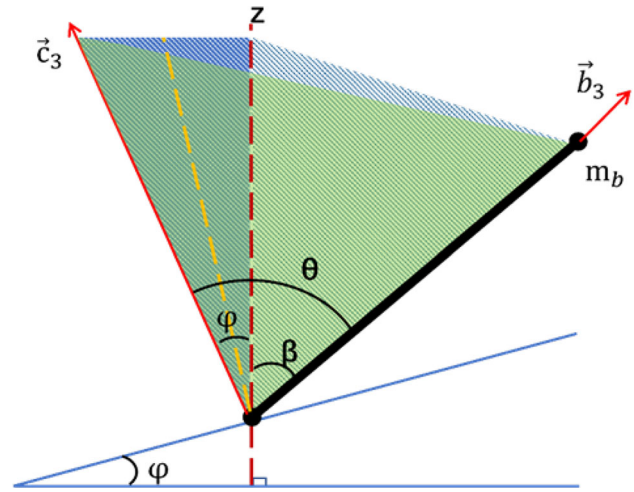

Fig. 5 The triangular pyramid constituted by $\vec{c}_1$-axis, $\vec{b}_3$-axis and the straight line z, where the $\vec{b}_3$-axis is not in the same plane with other lines, and the yellow line is the projection of $\vec{b}_3$-axis in the plane

With Eqs. (12–14) the potential energy can be written as:

$$\begin{aligned} V &= (m_B + 2m_W)gsin\varphi(r + x_C) \\ &+ m_B gl(sin\theta cos\psi sin\varphi + cos\theta cos\varphi). \end{aligned} \tag{15}$$

Therefore, the Lagrangian can be designed with Eqs. (11) and (15) as:

$$L = T - V = T_{trans} + T_{rot} - V, \tag{16}$$

which is the function of six generalized coordinates $q_1 = x_C$, $q_2 = y_C$, $q_3 = \theta$, $q_4 = \psi$, $q_5 = \gamma_L$,$q_6 = \gamma_R$. Then, the Lagrangian equation can be written as:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = Q_i + \sum_j \lambda_j a_{ij}, (i = 1\sim6, j = 1\sim3), \tag{17}$$

where Q_i denote the generalized force, λ_j are the Lagrange multipliers, and a_{ij} are the constraint equation multipliers.

Due to the assumption A4, the generalized forces corresponding to the generalized coordinates are only related to the torques T_L and T_R: $Q_1 = Q_2 = Q_4 = 0, Q_3 = -(Q_5 + Q_6), Q_5 = T_L, Q_6 = T_R$.

The a_{ij} can be calculated by the difference constraints Eq. (4):

$$\begin{aligned} a_{i1} &= \frac{\partial}{\partial \dot{q}_i}\left(\dot{x}_C cos\psi + \dot{y}_C sin\psi - \frac{d}{2}\dot{\psi} - r\dot{\gamma}_L\right), \\ a_{i2} &= \frac{\partial}{\partial \dot{q}_i}\left(\dot{x}_C cos\psi + \dot{y}_C sin\psi + \frac{d}{2}\dot{\psi} - r\dot{\gamma}_R\right), \\ a_{i3} &= \frac{\partial}{\partial \dot{q}_i}(\dot{x}_C sin\psi - \dot{y}_C cos\psi), \end{aligned} \tag{18}$$

Table 2 The Lagrange multipliers

a_{ij}	$j=1$	$j=2$	$j=3$
$i=1$	$\cos\psi$	$\cos\psi$	$\sin\psi$
$i=2$	$\sin\psi$	$\sin\psi$	$-\cos\psi$
$i=3$	0	0	0
$i=4$	$-\frac{d}{2}$	$\frac{d}{2}$	0
$i=5$	$-r$	0	0
$i=6$	0	$-r$	0

where the outcome is listed in Table. 2.

Substitute Eq. (18) into (17), then we have:

$$\begin{aligned}&(m_B+2m_W)\ddot{x}_C+m_Blcos\theta cos\psi\ddot{\theta}-m_Blsin\theta sin\psi\ddot{\psi}\\&-2m_Blcos\theta sin\psi\dot{\theta}\dot{\psi}-m_Blsin\theta cos\psi\left(\dot{\theta}^2+\dot{\psi}^2\right)\\&-(m_B+2m_W)gsin\varphi\\&=\lambda_1cos\psi+\lambda_2cos\psi+\lambda_3sin\psi,\end{aligned}\tag{19a}$$

$$\begin{aligned}&(m_B+2m_W)\ddot{y}_C+m_Blcos\theta sin\psi\ddot{\theta}+m_Blsin\theta cos\psi\ddot{\psi}\\&+2lcos\theta cos\psi\dot{\theta}\dot{\psi}-m_Blsin\theta sin\psi\left(\dot{\theta}^2+\dot{\psi}^2\right)\\&=\lambda_1sin\psi+\lambda_2sin\psi-\lambda_3cos\psi,\end{aligned}\tag{19b}$$

$$\begin{aligned}&(I_2+m_Bl^2)\ddot{\theta}+m_Blcos\theta(\ddot{x}_Ccos\psi+\ddot{y}_Csin\psi)\\&-(m_Bl^2+I_1-I_3)sin\theta cos\theta\dot{\psi}^2\\&+m_Bgl(cos\theta cos\psi sin\varphi-sin\theta cos\varphi)\\&=-(T_L+T_R),\end{aligned}\tag{19c}$$

$$\begin{aligned}&\left\{m_W\frac{d^2}{2}+(I_1-I_3)sin^2\theta+m_Bl^2sin^2\theta+I_3+2k\right\}\ddot{\psi}\\&+m_Blsin\theta(\dot{x}_Ccos\psi+\dot{y}_Csin\psi)\dot{\psi}\\&+2(I_1-I_3+m_Bl^2)sin\theta cos\theta\dot{\theta}\dot{\psi}\\&-m_Bglsin\theta sin\psi sin\varphi\\&=-\frac{d}{2}\lambda_1+\frac{d}{2}\lambda_2,\end{aligned}\tag{19d}$$

$$J\ddot{\gamma}_L=T_L-\lambda_1r,\tag{19e}$$

$$J\ddot{\gamma}_R=T_R-\lambda_2r.\tag{19f}$$

The Lagrange multipliers λ_1 and λ_2 can be solved from Eq. (19e) and (19f). λ_3 can be eliminated by Eq. (19a) multiply by $\sin\psi$ add to Eq. (19b) multiply by $\cos\psi$. Substitute Eqs. (5) and (6) into the Eq. (19), the standard model can be established by the generalized coordinates x, θ, ψ:

$$\begin{aligned}&\left(2m_w+m_b+\frac{2J}{r^2}\right)\ddot{x}+m_blcos\theta\ddot{\theta}-m_blsin\theta(\dot{\theta}^2+\dot{\psi}^2)\\&-(m_b+2m_w)g\cos\psi sin\varphi=\frac{T_L+T_R}{r},\\&(m_bl^2+I_2)\ddot{\theta}+m_blcos\theta\ddot{x}+(I_3-I_1-m_bl^2)sin\theta cos\theta\dot{\psi}^2\\&+m_bgl(cos\psi cos\theta sin\varphi-sin\theta cos\varphi)=-(T_L+T_R),\\&\left(I_3+\frac{d^2}{2}m_w+\frac{Jd^2}{2r^2}+2K+(m_bl^2+I_1-I_3)\sin^2\theta\right)\ddot{\psi}\\&+2(m_bl^2+l_1-l_3)sin\theta cos\theta\dot{\theta}\dot{\psi}+m_blsin\theta\dot{x}\dot{\psi}\\&-m_bglsin\theta sin\psi sin\varphi=\frac{d}{2r}(T_R-T_L).\end{aligned}\tag{20}$$

The key difference between this model and those built on a horizontal plane is the presence of zero-order terms created by the inclined terrain. When φ equals 0, the model becomes the commonly used TWIPR models, making the standard model Eq. (20) more extensive and general.

In addition, compared to the linear model or other models proposed in previous works [17] on sloped environments, the standard model includes more comprehensive first-order terms. The standard model will comprehensively elucidate the dynamic tendencies of TWIPRs on sloping terrain. There is a fact that has not been mentioned in the past works: when the TWIPR is not working along the direction of $\vec{n}_1$,which means the yaw angle ψ is not equal to 0, the angle between $\dot{x}$ and the horizontal plane is not equal to φ, but changes with the direction of $\dot{x}$ in the range of φ to $-\varphi$. This fact affects the target value of θ and the torque T_L,T_R. In particular, when the TWIPR is working along the direction of $\vec{n}_2$ at a uniform speed, the target value of θ is equal to 0. However, when the direction is $\vec{n}_1$, due to the influence of gravity, the target value depends on φ, which can be calculated from the first two equations in Eq. (20) as:

$$\begin{aligned}&\theta_d=arcsin\left(\frac{Hcos\psi}{\sqrt{1+tan^2\varphi\cos^2\psi}}\right)+arctan(tan\varphi cos\psi),\\&H=\frac{(m_b+2m_w)rtan\varphi}{m_bl}.\end{aligned}\tag{21}$$

Moreover, when the yaw rotation happens, the effect of nonlinear coupling terms cannot be ignored as same as the TWIPR working on a horizontal plane. It enables the standard model to more accurately describe the dynamical behavior of the TWIPR than previous models.

3 Control system design

In this section, SMC is applied to design the controllers. It will enable the TWIPR to maintain the balance and walk along the trajectory curve accurately on inclined terrain.

In order to design the control system, Eq. (20) is rewritten in the form as:

$$\begin{aligned}\ddot{x}&=A_1\dot{\theta}^2+B_1\dot{\psi}^2+C_1+D_1\tau_1,\\\ddot{\theta}&=A_2\dot{\theta}^2+B_2\dot{\psi}^2+C_2+D_2\tau_1,\\\ddot{\psi}&=A_3\dot{\theta}\dot{\psi}+B_3\dot{\psi}\dot{x}+C_3+D_3\tau_2,\end{aligned}\tag{22}$$

where $\tau_1 = T_L + T_R$ is the control of longitudinal motion and balance, $\tau_2 = T_R - T_L$ controlling the yaw motion and $A_k, B_k, C_k, D_k, k = 1-3$ are given in Appendix.

To track the control objective, the error states for control design are defined as follows:

$$\begin{aligned} &e_1 = x - x_d, e_2 = \dot{x} - \dot{x}_d, \\ &e_3 = \theta - \theta_d, e_4 = \dot{\theta} - \dot{\theta}_d, \\ &e_5 = \psi - \psi_d, e_6 = \dot{\psi} - \dot{\psi}_d. \end{aligned} \tag{23}$$

By Eqs. (22) and (23) the error differential equation can be written as follows:

$$\begin{aligned} &\dot{e}_1 = e_2, \\ &\dot{e}_2 = A_1\dot{\theta}^2 + B_1\dot{\psi}^2 + C_1 + D_1\tau_1 - \ddot{x}_d, \\ &\dot{e}_3 = e_4, \\ &\dot{e}_4 = A_2\dot{\theta}^2 + B_2\dot{\psi}^2 + C_2 + D_2\tau_1 - \ddot{\theta}_d, \\ &\dot{e}_5 = e_6, \\ &\dot{e}_6 = A_3\dot{\theta}\dot{\psi} + B_3\dot{\psi}\dot{x} + C_3 + D_3\tau_2 - \ddot{\psi}_d. \end{aligned} \tag{24}$$

3.1 Longitudinal control design and stability analysis

To apply the SMC, sliding surface is defined as:

$$s_1 = \dot{e}_1 + \lambda_1 e_1 = 0, \tag{25}$$

where λ_1 is positive. Then substitute first two equations of Eq. (24) into (25) with both sides of Eq. (25) is differentiated:

$$\dot{s}_1 = \ddot{e}_1 + \lambda_1\dot{e}_1 = A_1\dot{\theta}^2 + B_1\dot{\psi}^2 + C_1 + D_1\tau_1 - \ddot{x}_d + \lambda_1\dot{e}_1 = 0. \tag{26}$$

Typically, the approach law selected in sliding model control is $\dot{s} = -Ksgn(s)$. However, since sgn(s) is a discontinuous function, it can cause chattering. Therefore, we have chosen $\dot{s} = -k_i\frac{s}{1+s^2} - k_j s$. As this function is continuous, it reduces the discontinuity in the controller output, thereby mitigating the chattering phenomenon. Additionally, the term $-k_j s$ allows the adjustment of the convergence speed of s toward the sliding surface by selecting an appropriate value for k_j.

By adopting the approach law $\dot{s}_1 = -k_1\frac{s_1}{1+s_1^2} - k_2 s_1$, τ_1 can be designed as:

$$\tau_1 = -\frac{A_1}{D_1}\dot{\theta}^2 - \frac{B_1}{D_1}\dot{\psi}^2 - \frac{C_1}{D_1} + \frac{\ddot{x}_d}{D_1} - \frac{\lambda_1}{D_1}\dot{e}_1 - \frac{k_1}{D_1}\frac{s_1}{1+s_1^2} - \frac{k_2}{D_1}s_1, \tag{27}$$

where k_1, k_2 are positive.

To analysis the stability of the controller, the Lyapunov function is defined as follows:

$$V = \frac{1}{2}s_1^2. \tag{28}$$

Substitute Eq. (26) into Eq. (28) with both sides differentiated:

$$\begin{aligned} \dot{V} &= s_1\dot{s}_1 = (\ddot{e}_1 + \lambda_1\dot{e}_1)s_1 \\ &= (A_1\dot{\theta}^2 + B_1\dot{\psi}^2 + C_1 + D_1\tau_1 - \ddot{x}_d + \lambda_1\dot{e}_1)s_1. \end{aligned} \tag{29}$$

Then substitute Eq. (27) to Eq. (29):

$$\dot{V} = s_1\dot{s}_1 = -k_1\frac{s_1^2}{1+s_1^2} - k_2 s_1^2. \tag{30}$$

Obviously, Eq. (28) is positive definite and Eq. (30) is negative definite, which means the controller is effective. The stability analysis of control systems is basically the same, thus it will not be repeated in the next two parts.

3.2 Tilt angle control design

Both of the longitudinal motion and tilt motion are controlled by control input τ_1, which means the TWIPR is underactuated. Due to the longitudinal motion is coupled with the tilt motion, the zero-dynamics controller can be designed to solve the underactuated problem[18]. With $e_1 = 0$, which can derive $e_2 = 0$ and $\dot{e_2} = 0$, we can obtain the zero-dynamics system. In this situation, equation Eq. (27) changes to:

$$\tau_1 = -\frac{A_1}{D_1}\dot{\theta}^2 - \frac{B_1}{D_1}\dot{\psi}^2 - \frac{C_1}{D_1} + \frac{\ddot{x}_d}{D_1}, \tag{31}$$

where $\ddot{x}_d$ will be used for the input of tilt angle control. The sliding mode technique is used to derive the expression of $\ddot{x}_d$. The sliding surface is defined as:

$$s_2 = \dot{e}_3 + \lambda_2 e_3 = 0, \tag{32}$$

where λ_2 is positive. Then differentiate the sliding surface Eq. (32):

$$\dot{s}_2 = \ddot{e}_3 + \lambda_2\dot{e}_3 = A_2\dot{\theta}^2 + B_2\dot{\psi}^2 + C_2 + D_2\tau_1 - \ddot{\theta}_d + \lambda_2\dot{e}_3 = 0. \tag{33}$$

By substituting equation Eq. (31) into Eq. (33) and adopting the approach law $\dot{s}_2 = -k_3\frac{s_2}{1+s_2^2} - k_4 s_2$, $\ddot{x}_d$ can be designed

as:

$$\ddot{x}_d = \left(A_1 - \frac{D_1}{D_2}A_2\right)\dot{\theta}^2 + \left(B_1 - \frac{D_1}{D_2}B_2\right)\dot{\psi}^2 + \left(C_1 - \frac{D_1}{D_2}C_2\right) + \frac{D_1}{D_2}\left(\ddot{\theta}_d - \lambda_2\dot{e}_3 - k_3\frac{s_2}{1+s_2^2} - k_4 s_2\right), \tag{34}$$

where k_3 and k_4 are positive.

3.3 Rotational control design

The design process of rotational controller is the same as longitudinal control. The sliding surface is defined as:

$$s_3 = \dot{e}_5 + \lambda_3 e_5 = 0, \tag{35}$$

where λ_3 is positive. Then with the last two equations in Eq. (24), $\dot{s_3}$ can be calculated as:

$$\dot{s}_3 = \ddot{e}_5 + \lambda_3\dot{e}_5 = A_3\dot{\theta}\dot{\psi} + B_3\dot{\psi}\dot{x} + C_3 + D_3\tau_2 - \ddot{\psi}_d + \lambda_3\dot{e}_5 = 0. \tag{36}$$

By adopting the approach law $\dot{s}_3 = -k_5\frac{s_3}{1+s_3^2} - k_6 s_3$, the rotational controller is designed as:

$$\tau_2 = -\frac{A_3}{D_3}\dot{\theta}\dot{\psi} - \frac{B_3}{D_3}\dot{\psi}\dot{x} - \frac{C_3}{D_3} + \frac{1}{D_3}\ddot{\psi}_d - \frac{\lambda_3}{D_3}\dot{e}_5 - \frac{k_5}{D_3}\frac{s_3}{1+s_3^2} - \frac{k_6}{D_3}s_3, \tag{37}$$

where k_5 and k_6 are positive.

4 Simulation results

In this section, a simulation is conducted to evaluate the performance of the standard model and the three control laws based on the standard model. The parameters used in the simulation are listed in Table. 3. In addition, the initial conditions of the state variables are listed in Table. 4.

The target trajectory is a parabola, whose expression is $50y_{tt} = x_{tt}^2$ in the frame {N} with $y_{tt} = \frac{1}{2}t^2$ and $x_{tt} = 5t$, where t=5 s is the simulation time. The desired value of tilt angular is calculated by Eq. (21). To confirm the effectiveness of the standard model, two simulations have been performed. The first one designs the controller based on the standard model, while the second one uses the controller based on the linear model. Then the two proposed controllers are applied to the standard model.

Table 3 The parameters used in the simulation

Symbol	Value	Symbol	Value	Symbol	Value
m_B	3.99	I_1	0.068	k_1	10
m_W	1.064	I_2	0.043	k_2	10
K	0.043	I_3	0.044	k_3	15
J	0.009	d	0.4	k_4	15
g	9.81	r	0.3	k_5	10
φ	0.2	l	0.2	k_6	10
λ_1	10	λ_2	15	λ_3	10

Table 4 The initial conditions of the state variables

State variable	x	$\dot{x}$	θ	$\dot{\theta}$	ψ	$\dot{\psi}$
Value	0	0	0.3	0	0	0

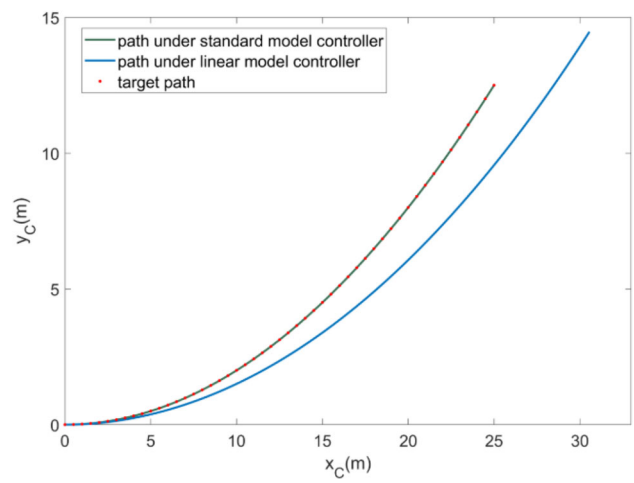

Fig. 6 Trajectory of the TWIPR on x-y plane. The red dots denote the target path while the green line is the path under the standard model controller and the blue line is the path under the linear model controller

Figure 6 shows the path under the controller based on the standard model, represented by the green line, closely overlapping with the target trajectory, suggesting that the standard model is suitable with no noticeable errors. In contrast, the blue line represents the trajectory of the TWIPR under a controller designed based on a linear model, with the error between this trajectory and the target trajectory increasing as it moves and the TWIPR's path significantly overshooting the target endpoint.

Figure 7 reveals the entire process along with the states at certain positions. Additionally, as the TWIPR follows the target path, both its position and yaw angle states are altered. Furthermore, the pitch angle decreases as the inclined angle of the TWIPR in yaw motion decreases. This implies that the standard model and its associated controller can address the underdrive problem of the TWIPR by facilitating the completion of pitch angle control of the body during lon-

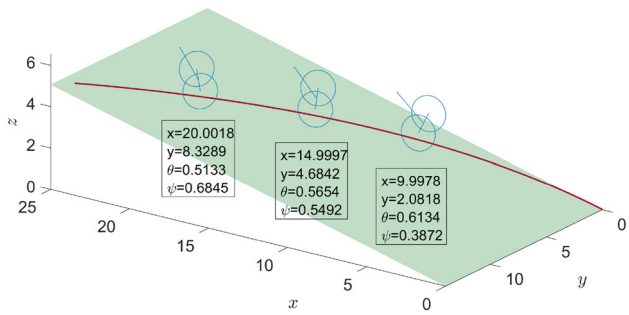

Fig. 7 The state of the TWIPR when target tracking on the inclined plane with the controller based on standard model

gitudinal control with a single input. This result is consistent with the analysis about Fig. 5 and around Eq. (21), which highlight how the system's dynamic behavior and control strategy inherently handle the complexities of underactuation. Ultimately, a well-designed model significantly aids in the controller design, ensuring that the goal-tracking task is effectively and accurately accomplished. The initial trajectory of the controller's linear model, depicted by the green line, aligns with the target trajectory at the onset. This finding is in line with earlier research asserting that the linear model path and the target path are congruent with slight yaw angles. However, as the TWIPR yaw angle increases, particularly when the target yaw angle surpasses 20 degrees (equivalent to a radian value beyond 0.34, near the initial state node in Fig. 7), the linear model's controller manifests considerable deviation.

The preceding analysis demonstrates that the control law derived from the standard model can precisely execute path tracking with significant yaw angles, whereas the linear model fails to do so. This is attributable to the linear model's portrayal of the TWIPR motion environment, which depicts a slope altering with the TWIPR velocity direction and leading to deviations from the predetermined slope in actual situations that affect the trajectory. The preceding analysis demonstrates that the control law derived from the standard model can precisely execute path tracking with significant yaw angles, whereas the linear model fails to do so. In contrast, the standard model comprehensively addresses this issue, thereby eliminating tracking errors.

With the intention of an intuitive description, the Euclidean distance e_{path} is used to evaluate the path tracking error.

$$e_{path} = \sqrt{(x_{tt} - x_c)^2 + (y_{tt} - y_c)^2}. \tag{38}$$

In Fig. 8a, initially, some path tracking errors appear during the adjustment of other states. However, the path tracking error then steadily decreases until it approaches zero. By contrast, although the path tracking error in Fig. 8b does not diverge, it fails to converge to zero. Table 5 provides more

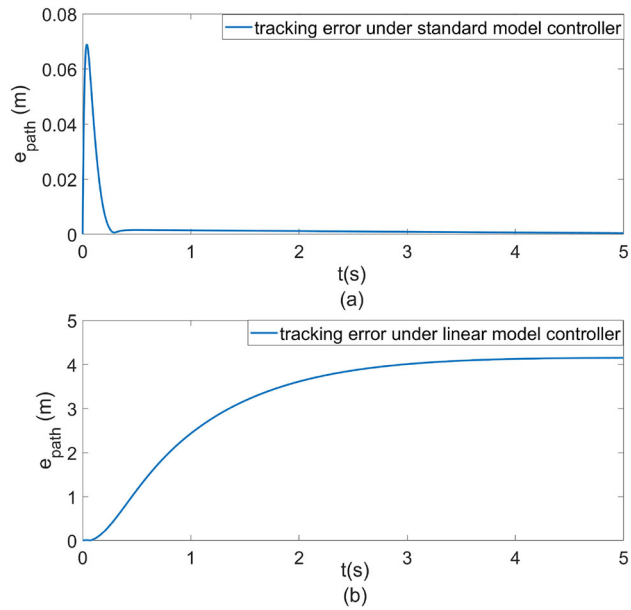

Fig. 8 The tracking error during TWIPR working on the inclined plane. (a) the controller based on standard model. (b) the controller based on linear model

Table 5 Results to establish the controller performance

path	standard model	linear model
final position	(24.9994, 12.5012)	(31.6576, 14.8614)
final error (m)	0.0013	7.0641
peak error (m)	0.1032	7.1346

detailed information that demonstrates the precision of the standard model. The results suggest that the controller following the standard model can reduce the final error to 0 gradually, with a peak error of 1.45% for the controller following the linear model.

Figure 9a displays that the pitch angle error converges to nearly zero within half a second, persistently decreasing thereafter. Note that the pitch angle also gradually decreases, as shown in Fig. 7. In contrast, Fig. 9 (b) indicates that both the pitch angle and the error still experience fluctuations. Specifically, the pitch angle error passes through zero and continually changes, causing a rapid decrease and subsequent increase in the path error before t=1 s in Fig. 8. The imprecision of the control law for the linear model is demonstrated by the connection of TWIPR pitch angle and longitudinal motion. A lack of accuracy in previous models accounts for this. Therefore, it is evident that the standard model yields higher accuracy, and the controller designed on this basis better enables the 3-DOF motion of TWIPRs on the inclined plane.

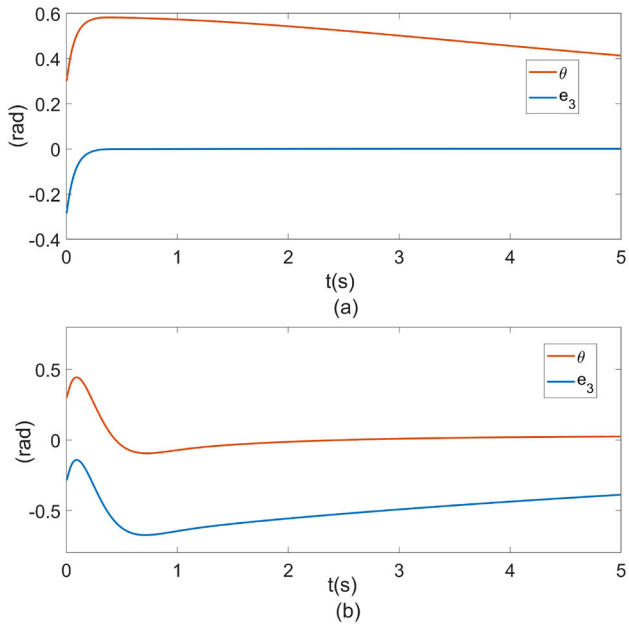

Fig. 9 The pitch angle error during TWIPR working on the inclined plane. (a) the controller based on standard model. (b) the controller based on linear model

5 Conclusion

This study introduces a standard 3-DOF nonlinear model for TWIPRs on inclined planes. This model addresses the limitations of traditional models where the associated control methods often neglect the complex dynamics of yaw motion and the interactions with forward motion and balance on inclined surfaces. By establishing the coordinate system on the inclined plane and focusing on the potential energy analysis, a standard model is developed using the Lagrangian method, and the pitch angle for the TWIPR when balanced on the inclined plane can be determined. The complex coupling terms in the equations, arising from potential energy and yaw motion, reveal the intricate dynamics of TWIPRs when operating on an inclined plane, which facilitates the development of an effective control method. Moreover, the accuracy of the model and controllers also is validated through analysis and simulations. Finally, simulations revealed that the control method using the standard model enables the TWIPR to achieve balance within 0.5 s. In contrast, the control method based on the linear model fails to maintain balance during simulations. In particular, for trajectory tracking, the standard model significantly reduces tracking errors, especially when the yaw angle is large whereas the linear model fails to complete the task. This result indicates that the proposed standard model is capable of completing trajectory tracking tasks for TWIPR's 3-DOF motion on inclined planes, highlighting its effectiveness and superiority over traditional models.

6 Appendix

$$a_{11} = 2m_w + m_b + \frac{2J}{r^2}, \quad a_{21} = a_{12} = m_b l cos\theta,$$
$$a_{22} = (m_b l^2 + I_2),$$
$$a_{33} = I_3 + \frac{d^2}{2}m_w + \frac{Jd^2}{2r^2} + 2K + (m_b l^2 + I_1 - I_3)sin^2\theta,$$
$$f_1 = -m_b l sin\theta, \quad f_2 = (I_3 - I_1 - m_b l^2)sin\theta cos\theta,$$
$$f_{31} = 2(m_b l^2 + l_1 - l_3)sin\theta cos\theta, \quad f_{32} = -f_1,$$
$$g_1 = -(m_b + 2m_w)g cos\psi sin\varphi,$$
$$g_2 = m_b g l(cos\psi cos\theta sin\varphi - sin\theta cos\varphi),$$
$$g_3 = -m_b g l sin\theta sin\psi sin\varphi,$$
$$A_1 = \frac{-a_{22}f_1}{a_{11}a_{22} - a_{12}^2}, A_2 = \frac{a_{12}f_1}{a_{11}a_{22} - a_{12}^2}, A_3 = -\frac{f_{31}}{a_{33}},$$
$$B_1 = \frac{a_{12}f_2 - a_{22}f_1}{a_{11}a_{22} - a_{12}^2}, B_2 = \frac{a_{12}f_1 - a_{11}f_2}{a_{11}a_{22} - a_{12}^2}, B_3 = -\frac{f_{32}}{a_{33}},$$
$$C_1 = \frac{a_{12}g_2 - a_{22}g_1}{a_{11}a_{22} - a_{12}^2}, C_2 = \frac{a_{12}g_1 - a_{11}g_2}{a_{11}a_{22} - a_{12}^2}, C_3 = -\frac{g_3}{a_{33}},$$
$$D_1 = \frac{\frac{a_{22}}{r} + a_{12}}{a_{11}a_{22} - a_{12}^2}, D_2 = \frac{-\frac{a_{12}}{r} - a_{11}}{a_{11}a_{22} - a_{12}^2}, D_3 = \frac{d}{2ra_{33}}.$$

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Author Contributions Z. Sun conducted the theory and stability analysis. R. Dong performed the simulations. R. Dong and N. Zhao prepared all figures and wrote the initial draft. Z. Sun reviewed the manuscript. Z. Sun and N. Zhao validated the research outcomes.

Data Availability No datasets were generated or analyzed during the current study.

Code Availability The code used in this study is available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare that they have no Conflict of interest.

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